

Abdalla Samir Khalifa

Java Backend Developer

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Summary

Backend Software Engineer specializing in Java, Spring Boot, and PostgreSQL, focused on building well-architected, testable backend systems. Comfortable across the full development lifecycle, from UML system analysis to Docker deployment. Solid foundation in data structures and algorithmic problem-solving, reinforced through hands-on implementation across Java, C#, and C++.

Projects

Food Delivery Backend Platform (Team Work) 🔗

- Collaborated on a DDD-inspired architecture, with rich domain entities managing their own state mutations and services following aggregate-root delegation, and refactored existing code for clarity and maintainability.
- Designed and implemented Chain of Responsibility pipeline for order placement – cart validation, open-time check, item availability, order finalization, and payment – executing within a single transaction.
- Designed and implemented the platform's security model: stateless JWT authentication, OTP email verification, and role-based access control across Customer/Admin roles.
- Owned core infrastructure: Docker containerization, PostgreSQL schema and JPA entity design, and integration testing via Singleton Testcontainers against a real PostgreSQL instance – no in-memory substitutes.

E-Commerce Backend Platform 🔗

- Designed and built a Spring Boot 3 REST API solo, covering catalog, cart, and checkout, following a full UML system-analysis phase before implementation.
- Implemented JWT authentication and role-based access control for secured endpoints, with HATEOAS-linked responses for discoverable, self-documenting APIs.
- Containerized the application with Docker Compose and applied targeted foreign-key indexing on high-selectivity columns to improve query performance.

Database Design & Optimization 🔗

- Designed and optimized a PostgreSQL schema for 50M+ synthetic records generated via shell script, reducing query latency by **98.8%** (613ms to 7ms) through indexing, query rewriting, and execution plan analysis.
- Implemented a denormalization layer that improved analytical query performance by **99.9%** (16sec to 12ms).

Skills

Programming Languages: Java, C#, C++, SQL

Backend Development: Spring Boot, Spring Web, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, JWT, Swagger/OpenAPI.

Databases: PostgreSQL, Database Design, Performance Tuning.

Testing: JUnit, Mockito, Testcontainers, Unit Testing, Integration Testing.

DevOps & Tools: Docker, Git, Maven, Linux.

AI-Assisted Development: Claude, Gemini.

Software Engineering: UML, Data Structures and Algorithms, OOP, Design Patterns, Domain-Driven Design (basic), SOLID Principles, Code Refactoring, System Design, Agile, Problem Solving.

Frontend Development (basic): Angular, JavaScript, TypeScript, HTML/CSS.

Certifications

DeepLearning.AI

May 2026

Introduction to Generative AI for Software Development 🔗

Education

Higher Institute of Management Sciences

October 2021 – June 2025

Bachelor of Management Information Systems

Very Good / GPA: 3.02/4.0

Languages

Arabic: Native

English: Professional Working Proficiency